You've now seen some basics for individual query optimization. As your queries become more complicated, you want to reach for more advanced techniques.

Advanced Query Support In Active Record

Subquery Expressions[11] or *Subqueries* are what PostgreSQL calls multiple SQL queries combined into one statement. The general pattern is that an outer query uses a second inner query as a way to filter rows down. Filtering is done using a WHERE clause or IN clause. Subqueries are common in SQL and recently support was added to Active Record.

You'll adapt an example from the PostgreSQL Tutorial on Subqueries[12] to Rideshare creating an analytical type of query to show what's possible in a single statement. You'll first write the query in SQL and then convert it to Active Record. The query has these goals:

- Find the average number of Trips for all Drivers
- Find Drivers with more Trips than average
- Order above average Drivers by Trip count in descending order

To find the average in SQL, use the AVG() aggregate function. If you'd been working in bin/rails console, go back to psql for these since you're running SQL queries.

sql/select_avg_trips_count.sql

```
SELECT AVG(trips_count) FROM users
WHERE type = 'Driver';
```

Use the average Trip Count as *input* to a WHERE clause outer query, by making the Trip Count query an inner query or subquery. Modify the query below to find drivers above and below the average.

sql/trips_count_subquery.sql

```
SELECT
  id, trips_count
FROM users
WHERE type = 'Driver'
AND trips_count > (
  -- the earlier AVG(trips) query
  SELECT AVG(trips_count) FROM users
  WHERE type = 'Driver'
)
ORDER BY trips_count
DESC LIMIT 5;
```

11. https://www.postgresql.org/docs/current/functions-subquery.html
12. https://www.postgresqltutorial.com/postgresql-tutorial/postgresql-subquery/